LECTURE 9: INTRODUCTION TO UNCONFOUNDED TREATMENT ASSIGNMENT

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OUTLINE

1. Introduction and Notation
2. Description of Applications and Summary Statistics
3. Assumptions
4. Concerns with Linear Regression Methods
5. Design: Assessing and Ensuring Overlap
6. Assessing Unconfoundedness
7. Estimation of Average Treatment Effects (Lecture 11)

INTRODUCTION

- Two applications where unconfoundedness/selection-on-observables /exogeneity of treatment may be reasonable.
- How do we go about analyzing these data?
  - Summary Statistics
  - Design: ensuring overlap
  - Assessing plausibility of unconfoundedness (should really be done before estimation of average treatment effects)
  - Estimates of average treatment effects (lecture 11)

NOTATION

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- Treatment indicator: $W_i \in \{0, 1\}$ (sometimes $W_i \in \{C, T\}$)
- Potential Outcomes $Y_i(C), Y_i(T)$, unit-causal effect $\tau_i = Y_i(T) - Y_i(C)$
- Covariates/pre-treatment variables X_i, not affected by treatment.
- Realized/Observed Outcome:

$$Y_i^{\text{obs}} = W_i \cdot Y_i(T) + (1 - W_i) \cdot Y_i(C)$$

- $N_C = \sum_i (1 - W_i)$ number of controls, $N_T = \sum_i W_i$ number of treated units.
- $\overline{Y}_C^{\text{obs}}$ and $\overline{Y}_T^{\text{obs}}$ are average (realized/observed) outcomes for controls and treated. $\overline{Y}(C)$ and $\overline{Y}(T)$ are averages of potential outcomes

$$\overline{Y}_C^{\text{obs}} \equiv \frac{1}{N_C} \sum_{i=1}^{N} (1 - W_i) Y_i = \frac{1}{N_C} \sum_{i=1}^{N} (1 - W_i) Y_i(C) \qquad \overline{Y}(C) \equiv \frac{1}{N} \sum_{i=1}^{N} Y_i(C)$$

APPLICATION I: IMBENS-RUBIN-SACERDOTE LOTTERY STUDY

- IRS were interested in estimating the effect of unearned income on economic behavior including effects on labor supply, consumption and savings.
- Those effects are important to understand and improve design of social security programs.
- IRS surveyed individuals who had played and won large sums of money in the lottery ("winners"). As a comparison group they collected data on a second set of individuals who also played the lottery but who had not won big prizes (the "losers") .

APPLICATION I: IMBENS-RUBIN-SACERDOTE LOTTERY STUDY

- $N_T = 259$ winners and $N_C = 237$ losers in the sample of $N = 496$ lottery players.
- pre-treatment variables: We know the year individuals played the lottery, the number of tickets they typically bought, age, sex, education, and their social security earnings for the six years preceeding their winning.
- What else should we have asked for?
- We first present
  - averages and standard deviations for the full sample,
  - t-statistics for the null hypothesis of equal means in the two groups,
  - normalized differences in the means.

NORMALIZED DIFFERENCE IN COVARIATES

- Define

$$\overline{X}_{\mathrm{C}} \equiv \frac{1}{N_c} \sum_{i:W_i=0} X_i \qquad S^2_{\mathrm{C}} \equiv= \frac{1}{N_{\mathrm{C}}-1} \sum_{i:W_i=0} \left(X_i - \overline{X}_{\mathrm{C}}\right)^2$$

$$\overline{X}_{\mathrm{T}} \equiv \frac{1}{N_{\mathrm{T}}} \sum_{i:W_i=1} X_i \qquad S^2_{\mathrm{T}} \equiv \frac{1}{N_{\mathrm{T}}-1} \sum_{i:W_i=1} \left(X_i - \overline{X}_{\mathrm{T}}\right)^2$$

- Normalized difference:

$$\mathrm{nd} \equiv \frac{\overline{X}_t - \overline{X}_c}{\sqrt{(S^2_c + S^2_t)/2}}$$

- More relevant for assessing balance than the t-statistic

$$t \equiv \frac{\overline{X}_{\mathrm{T}} - \overline{X}_{\mathrm{C}}}{\sqrt{S^2_{\mathrm{C}}/N_{\mathrm{C}} + S^2_{\mathrm{T}}/N_{\mathrm{T}}}}$$

SUMMARY STATISTICS IRS LOTTERY SAMPLE (NO OUTCOMES)

Variable	All		Losers	Winners		
	N=496		N_T=259	N_C=237		Norm.
	mean	(s.d.)	mean	mean	[t-stat]	Dif.
Year Won	6.23	1.18	6.38	6.06	-3.0	-0.27
# Tickets	3.33	2.86	2.19	4.57	9.9	0.90
Age	50.2	13.7	53.2	47.0	-5.2	-0.47
Male	0.63	0.48	0.67	0.58	-2.1	-0.19
Education	13.73	2.20	14.43	12.97	-7.8	-0.70
Working Then	0.78	0.41	0.77	0.80	0.9	0.08
Earn Y -6	13.8	13.4	15.6	12.0	-3.1	-0.27
⋮						
Earn Y -1	16.3	15.7	18.0	14.5	-2.5	-0.23
Pos Earn Y-6	0.69	0.46	0.69	0.70	0.3	0.03
Pos Earn Y-1	0.71	0.45	0.69	0.74	1.2	0.10

PRE-TREATMENT DIFFERENCES FOR THE IMBENS-RUBIN-SACERDOTE LOTTERY STUDY

- There are substantial differences between the two groups, including in pre-winning earnings (18.0 *versus* 14.5) and the number of lottery tickets bought (2.19 *versus* 4.57 with a normalized difference of 0.90).
- This suggests that simple regression methods will not necessarily be adequate for removing the biases associated with the differences in covariates.
- Where do these differences come from?
  - People buying more tickets are more likely to win.
  - Nonresponse may differ by prize and individual characteristics (including labor income)
- Is unconfoundedness plausible?

APPLICATION II: LALONDE STUDY

- Lalonde (1986) analyzed data from a randomized experiment designed to evaluate the effect of a labor market program, the National Supported Work (NSW) program.
- Men with poor labor market histories.
- Randomized evaluation shows average effect on earnings $\approx$ \$2,000 (Substantial for participants).
- Lalonde: could we have estimated this without randomized experiment?

LALONDE EVALUATION OF THE EFFECTIVENESS OF OBSERVATIONAL METHODS TO RECOVER CAUSAL EFFECTS:

- Put aside experimental control group.
- Constructs comparison sample from public use data set: Current Population Survey (CPS), and Panel Study of Income Dynamics (PSID).
- Estimate average effect using CPS (or PSID) and trainees, as observational study.
- Is estimated effect close to experimental estimate?
- Lalonde concludes that nonexperimental evaluations are not credible. Influential conclusion in policy circles.

SUMMARY STATISTICS LALONDE STUDY

	All		CPS Controls $N_C = 15,992$	Trainees $N_T = 185$		Norm.
	mean	(s.d.)	mean	mean	t-stat	Diff.
Black	0.08	0.27	0.07	0.84	28.6	2.4
Hisp	0.07	0.26	0.07	0.06	-0.7	-0.1
Age	33.1	11.0	33.2	25.8	-13.9	-0.8
Married	0.71	0.46	0.71	0.19	-18.0	-1.2
Nodegree	0.30	0.46	0.30	0.71	12.2	0.9
Education	12.0	2.9	12.0	10.4	-11.2	-0.7
E’74	13.9	9.6	14.0	2.1	-32.5	-1.6
U’74	0.13	0.33	0.12	0.71	17.5	1.5
E’75	13.5	9.3	13.7	1.5	-48.9	-1.7
U’75	0.11	0.32	0.11	0.60	13.6	1.2

ASSUMPTIONS: UNCONFOUNDEDNESS

- Unconfoundedness

$$W_i \perp\!\!\!\perp \Big(Y_i(0), Y_i(1)\Big) \Big| X_i.$$

- This form due to Rosenbaum and Rubin (1983). Sometimes referred to as selection on observables, or exogeneity but those terms are not quite so well defined.
- Suppose

$$Y_i(0) = \alpha + \beta' X_i + \varepsilon_i, \qquad \text{and} \;\; Y_i(1) = Y_i(0) + \tau,$$

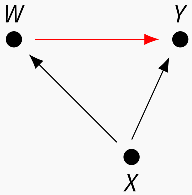
- Then

$$Y_i = \alpha + \tau \cdot W_i + \beta' X_i + \varepsilon_i,$$

- Then unconfoundedness $\Longleftrightarrow \varepsilon_i \perp W_i | X_i$ (exogeneity)

DAG CORRESPONDING TO UNCNFOUNDEDNESS

- (X may be vector-valued)



- (leaving out the unobserved nodes that only affect a single observed node).
- Structural Equation Model for this graph:

$$X = f_1(\varepsilon_1) \qquad W = f_2(X, \varepsilon_2) \qquad Y = f_3(X, W, \varepsilon_3)$$

MOTIVATION FOR UNCONFOUNDEDNESS

1. Descriptive statistics. After simple difference in mean outcomes by treatment status, compare average outcomes adjusted for covariates.
2. Unconfoundedness follows from some economic models: Suppose individuals choose treatment w to max expected utility, equal to outcome minus cost c_i, $Y_i(w) - c_i \cdot w$, given a set of covariates X:

$$W_i = \text{argmax}_w \, \mathbb{E}[Y_i(w)|X_i] - c_i \cdot w.$$

   Suppose that costs vary between individuals, independent of potential outcomes, $c_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$. Then
   - choices will vary between individuals with the same covariates,
   - conditional on the covariates X the choice is independent of the potential outcomes.
3. Alternative: bounds (e.g., Manski, 1990)
4. Unconfoundedness is not testable

ASSUMPTIONS: OVERLAP

- Overlap in covariate distributions

$$0 < \mathrm{pr}(W_i = 1 | X_i) < 1.$$

- $\Longrightarrow$ for all X there are in large samples treated and control units.
- This assumption is in principle testable: one can estimate the propensity score

$$e(x) = \mathrm{pr}(W_i = 1 | X_i = x)$$

  and assess whether it gets close to zero.
- Unconfoundedness and overlap combined: Strong Ignorability (Rosenbaum and Rubin, 1983)
- This is what we will work with for these two applications.
  - Is this reasonable?
  - Can we assess how reasonable this is?

IDENTIFICATION GIVEN UNCONFOUNDEDNESS AND OVERLAP

- By definition,

$$\tau(X) \equiv \mathbb{E}[Y_i(1) - Y_i(0)|X_i = x] = \mathbb{E}[Y_i(1)|X_i = x] - \mathbb{E}[Y_i(0)|X_i = x]$$

- By unconfoundedness this is equal to

$$\mathbb{E}[Y_i(1)|W_i = 1, X_i = x] - \mathbb{E}[Y_i(0)|W_i = 0, X_i = x]$$

- By definition this is equal to

$$\mathbb{E}[Y_i|W_i = 1, X_i = x] - \mathbb{E}[Y_i|W_i = 0, X_i = x].$$

- By the overlap assumption we can estimate both terms.
- Then

$$\tau = \mathbb{E}[\tau(X_i)].$$

RECAP/ROADMAP

- Given sample (W_i, Y_i, X_i), interested in estimating average adjusted difference

$$\tau \equiv \mathbb{E}\left[\mathbb{E}\left[Y_i | W_i = 1, X_i\right] - \mathbb{E}\left[Y_i | W_i = 0, X_i\right]\right]$$

- Under unconfoundedness this is equal to average causal effect $\mathbb{E}[Y_i(\mathrm{T}) - Y_i(\mathrm{C})]$ but adjusted difference is well-defined (but not necessarily interesting!) without unconfoundedness.
- Questions in today's lecture:
  - Concerns with regression estimators
  - Concerns with overlap
  - "Validation" of unconfoundedness assumption
- Lecture 10: How well can we estimate adjusted difference?
- Lecture 11: How do we estimate adjusted difference?

CONCERN WITH OVERLAP

- There are two potential concerns with lack of overlap:
  - Sensitivity to modelling assumptions: if there is only limited overlap, estimates may be very sensitive to small changes in the specification that we don't really know how to make.
  - Lack of precision: if there is limited overlap, we may not be able to estimate the average effect precisely.
    - ★ For values of X_i where we have very few control units or very few treated units we cannot estimate the average treatment effect well.
    - ★ If there are such values that are common in the sample, we cannot estimate the overall average effect precisely.
- Need to assess and account for lack of overlap.

CONCERN WITH REGRESSION ESTIMATORS

- Distinct from plausibility of unconfoundedness assumption.
- Regression estimators are the most widely used methods for estimating treatment effects
- Sometimes simple linear regression:

$$Y_i^{\text{obs}} = \beta_0 + \tau \cdot W_i + \beta_1 \cdot X_i + \varepsilon$$

- Sometimes allowing for interactions with pre-treatment variables:

$$Y_i^{\text{obs}} = \beta_0 + \tau \cdot W_i + \beta_1 \cdot X_i + \beta_2 \cdot (X_i - \overline{X}) \cdot W_i + \varepsilon$$

- What is the problem with linear regression here?

CONCERN WITH REGRESSION ESTIMATORS

- Suppose the regression model is:

$$Y_i(w) = \beta_{w0} + \beta_{w1} \cdot X_i + \varepsilon_{wi} \qquad w = \mathrm{C}, \mathrm{T}$$

- Suppose we are interested in the population average for the treated, τ_t^P:

$$\tau_t^P = \mathbb{E}[Y_i(\mathrm{T}) - Y_i(\mathrm{C})|W_i = 1] = \mathbb{E}[Y_i|W_i = 1] - \mathbb{E}[\mathbb{E}[Y_i|W_i = 0, X_i]|W_i = 1]$$

- This is estimated as $\overline{Y}_\mathrm{T} - \overline{Y}_\mathrm{C} - \hat{\beta}_{\mathrm{C}1} \cdot \left(\overline{X}_\mathrm{T} - \overline{X}_\mathrm{C}\right)$
- The problem is that if the difference $\overline{X}_t - \overline{X}_c$ is substantial, we do a lot of extrapolation using this estimator.
- Regression is OK with experimental data, (so that $\overline{X}_\mathrm{T} - \overline{X}_\mathrm{C}$ is zero in expectation) but not if $\overline{X}_\mathrm{T} - \overline{X}_\mathrm{C}$ is substantial.
- This is why looking at normalized differences in covariates is useful as indication of potential problems.

SENSITIVITY OF REGRESSION ESTIMATES FOR THE LALONDE DATA

- Four regression estimates, all using single covariate, earnings in 1975 (X_i), in thousands of dollars (normalized difference -1.3)
  - $\hat{\tau} = -8.5$ (s.e. 0.7) $\quad Y_i(w) = \beta_{0w} + \varepsilon_{iw}$
  - $\hat{\tau} = -0.1$ (s.e. 0.5) $\quad Y_i(w) = \beta_{0w} + \beta_{1w} \cdot X_i + \varepsilon_{iw}$
  - $\hat{\tau} = 1.2$ (s.e. 0.6) $\quad Y_i(w) = \beta_{0w} + \beta_{1w} \cdot \log(X_i + 1) + \varepsilon_{iw}$
  - $\hat{\tau} = 0.1$ (s.e. 0.5) $\quad Y_i(w) = \beta_{0w} + \beta_{1w} \cdot X_i + \beta_{2w} \cdot X_i \cdot X_i + \varepsilon_{iw}$
- Estimates vary widely by specification

DESIGN: ASSESSING AND ENSURING OVERLAP

- Lack of overlap in covariate distributions makes many conventional estimators (*e.g.*, regression) sensitive to minor changes in specification and leads to imprecise estimates
- Overlap of the covariate distributions is most easily assessed using the propensity score.
- Note: it is not sufficient that there is overlap in the marginal distributions for each of the covariates separately.
- Calculate means for the log odds ratios for the propensity score for the two groups. If more than a standard deviation apart, regression can be very misleading as it relies a lot on extrapolation.

ASSESSING OVERLAP

- If the covariate distributions differ, it shows up in the propensity score:

$$e(x) \equiv \mathrm{pr}(W_i = 1 | X_i = x) \qquad p \equiv \mathrm{pr}(W_i = 1)$$

- Then:

$$\mathbb{V}(e(X_i)) = p^2(1 - p)^2 \mathbb{E}\left[\left(\frac{f_{\mathrm{T}}(X_i) - f_{\mathrm{C}}(X_i)}{p\, f_{\mathrm{T}}(X_i + (1 - p)\, f_{\mathrm{C}}(X_i)}\right)^2\right]$$

$$= p(1 - p)\left(\mathbb{E}[e(X_i)|W_i = 1] - \mathbb{E}[e(X_i)|W_i = 0]\right)$$

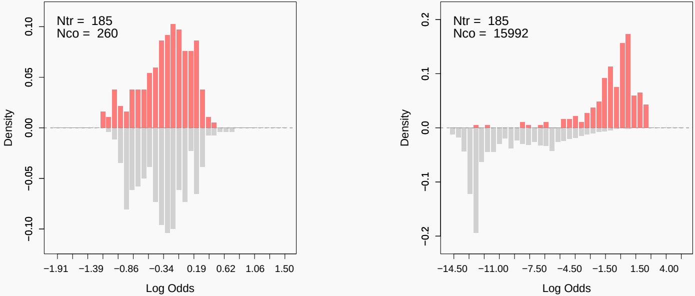
SUMMARY STATISTICS FOR LALONDE DATA: EXPERIMENTAL AND OBSERVATIONAL DATA

	Experimental			Non-exp		
	Means		norm.	Means		norm.
	contr	treat	diff.	contr	treat	diff.
pscore	0.39	0.46	(0.54)	0.01	0.41	(1.94)
log odds ratio	-0.49	-0.18	(0.53)	-10.29	-0.76	(2.51)

- log odds ratio is $\ln(e(x)/(1-e(x)))$, where $e(x) = \text{pr}(W_i = 1|X_i = x)$ is the propensity score.
- For nonexperimental sample 92% of controls has p-score ≤ 0.01, *versus* 6% of treated.
- Big differences in propensity score for treated and controls in nonexperimental sample.

ASSESSING OVERLAP WITH THE PROPENSITY SCORE

- From Imbens and Xu (2024): the distribution of the log odds ratio for the LaLonde data, experimental and CPS.



TWO OPTIONS IN CASE WITH LIMITED OVERLAP

- Under unconfoundedness, if overlap in covariates between treated and controls is limited, the population average treatment effect is difficult to estimate.
  - Create matched sample to estimate average effect on treated.
  - Create subsample with overlap.

OPTION I FOR CREATING BALANCED SAMPLE: MATCHED SAMPLE

- Focus on average effect for treated. (e.g., Lalonde data):

$$\tau_t^P \equiv \mathbb{E}[Y_i(1) - Y_i(0)|W_i = 1]$$

- Note: the focus is not on estimation of treatment effects yet. We first create a sample with more balance.
- Match on propensity score (better here than matching on all covariates).
  - First match the treated observation with the highest value of the propensity score to the closest control. (most difficult to find a match for)
  - Go down the list of treated units till all are matched to a unique control (no replacement)
- Now analyse the sample ignoring the matching step. (do regression, do matching, do propensity score analysis)

MATCHING TO IMPROVE BALANCE

- We first estimate the propensity score, using the 185 trainees and 15,992 CPS controls.
- Then we order the 185 trainees by the estimated propensity score, the largest first.
- Then, going down the list of trainees, we match each to the nearest CPS control, in terms of the estimated propensity score, without replacement.
- Now we have a new data set with 185 treated and 185 matched CPS controls.
- Balance is much improved.

THE LALONDE DATA: NORMALIZED DIFFERENCES AFTER MATCHING

	All	Matched Sample Covariate	Matched Sample Propensity	Ratio of Propensity / All
Black	3.83	0.00	-0.04	-0.01
Hispanic	-0.10	0.00	-0.05	0.45
Age	-0.04	-0.00	-0.01	0.13
Married	-1.45	-0.16	-0.02	0.01
Nodegree	0.99	0.10	0.11	0.11
Education	-0.14	-0.00	-0.00	0.03
E’74	-0.10	0.00	-0.01	0.12
U’74	1.88	0.11	0.09	0.05
E’75	-0.13	-0.01	-0.01	0.07
U’75	1.45	0.04	0.11	0.08

- Matching on propensity score generally better than on individual covariates with many covariates.

DESIGN OPTION II: TRIMMING THE SAMPLE

- Traditional Estimand $\tau^P = \mathbb{E}[Y_i(1) - Y_i(0)] = E[\tau(X_i)]$ (Population Average Treatment Effect)
- Problem: τ^P can be difficult to estimate when there are values $x \in \mathbb{X}$ such that $e(x)$ close to zero or one.
- In that case:

  estimates are imprecise (high variance)estimates are sensitive to small changes in specification

SYSTEMATIC SOLUTION

- Focus on new estimand.

- For a subset $\mathcal{A}$ of the covariate space, focus on

$$\tau_C(\mathcal{A}) = \frac{1}{\#\mathcal{A}} \sum_{i:X_i \in \mathcal{A}} \tau(X_i)$$

- (Subpopulation Average Treatment Effect)

- Drop units with $X_i \notin \mathcal{A}$

- Question: How do we choose $\mathcal{A}$?

- We formalize this as:
  - Which set $\mathcal{A}$ minimizes the variance of $\tau_C(\mathcal{A})$?
  - Shrinking $\mathcal{A}$ decreases the number of units in $\mathcal{A}$ and so increases the variance
  - Expanding

ILLUSTRATION: BINARY X CASE

- $X \in \{f, m\}$
- N_x is sample size for the subsample with $X = x$
- $q = N_m/N$ be the population share of $X = m$ units.
- τ_x is average treatment effect conditional on the covariate
- $\tau = q \cdot \tau_m + (1 - q) \cdot \tau_f$.
- N_{wx} is number of observations with covariate $X_i = x$ and treatment $w \in \{\mathrm{C}, \mathrm{T}\}$.
- $e_x = N_{x\mathrm{T}}/N_x$ is propensity score for $x = f, m$.
- $\overline{Y}_{\mathrm{C}x} = \sum_{i:W_i=0,X_i=x} Y_i/N_{\mathrm{C}x}, \overline{Y}_{\mathrm{T}x} = \sum_{i:W_i=1,X_i=x} Y_i/N_{\mathrm{T}x}$
- Assume that the variance of $Y_i(w)$ given $X_i = x$ is σ^2 for all x (makes procedure feasible).

ILLUSTRATION: BINARY X CASE

- Then

$$\hat{\tau}_x = \overline{Y}_{\mathrm{T}x} - \overline{Y}_{\mathrm{C}x}$$

with variances

$$V(\hat{\tau}_f) = \frac{\sigma^2}{N \cdot (1-q)} \cdot \frac{1}{e_f \cdot (1-e_f)}, \qquad V(\hat{\tau}_m) = \frac{\sigma^2}{N \cdot q} \cdot \frac{1}{e_m \cdot (1-e_m)},$$

- The estimator for the population average treatment effect is

$$\hat{\tau} = q \cdot \hat{\tau}_m + (1-q) \cdot \hat{\tau}_f.$$

with variance

$$V(\hat{\tau}) = \frac{\sigma^2}{N} \cdot \mathbb{E}\left[\frac{1}{e_X \cdot (1-e_X)}\right].$$

ILLUSTRATION: BINARY X CASE

- Define $V = \min(V(\hat\tau), V(\hat\tau_f), V(\hat\tau_m))$. Then

$$V = \begin{cases} V(\hat\tau_f) & \text{if} & & \frac{e_m(1-e_m)}{e_f(1-e_f)} & \leq \frac{1-q}{2-q}, \\ V(\hat\tau) & \text{if} & \frac{1-q}{2-q} \leq & \frac{e_m(1-e_m)}{e_f(1-e_f)} & \leq \frac{1+q}{q}, \\ V(\hat\tau_m) & \text{if} & \frac{1+q}{q} \leq & \frac{e_m(1-e_m)}{e_f(1-e_f)}. & \end{cases}$$

- If e_f is close to zero or one, we may be better of focusing on τ_m.
- If e_m is close to zero or one, we may be better of focusing on τ_f.
- If both are far from zero and one, we can focus on τ.
- Better off here means that we can estimate the corresponding estimand more accurately.

GENERAL CASE

- Optimal set:

$$\mathcal{A} = \left\{ x \in \mathbb{X} \,\middle|\, \alpha < e(x) < 1 - \alpha \right\},$$

where optimal α depends on the distribution of the propensity score in the sample. (See Crump, Hotz, Imbens, Mitnik, Biometrika 2009)

- Formally the optimal α solves:

$$\alpha = 2 \cdot \mathbb{E}\left[\frac{1}{e(X) \cdot (1 - e(X))} \,\middle|\, \frac{1}{e(X) \cdot (1 - e(X))} < \alpha \right].$$

- Approximate solution:

$$\mathcal{A}^* = \left\{ x \in \mathbb{X} \,\middle|\, 0.1 < e(x) < 0.9 \right\}$$

- Practical recommendation: drop units with a propensity score below 0.1 or above 0.9.

THE OPTIMAL SUBSAMPLE FOR THE LOTTERY DATA

- Based on the estimated propensity score we calculate the optimal value for the threshold α to be 0.0891.
- Out of the 259 losers 82 have an estimated propensity score less than $\alpha = 0.0891$, and 5 have an estimated propensity score greater than $1 - \alpha = 0.9109$.
- This leaves us with 172 losers in the selected sample.
- Of the 237 winners, 4 have an estimated propensity score less than $\alpha = 0.0891$, and 82 have an estimated propensity score greater than 0.9109, leaving us with 151 treated observations.
- This leaves us with 323 individuals in the selected (trimmed) sample.

SAMPLE SIZES FOR SELECTED SUBSAMPLES WITH THE PROPENSITY SCORE BETWEEN α AND $1 - \alpha$ (α =0.0891)

	low	middle	high	All
	$e(x) < \alpha$	$\alpha \leq e(X) \leq 1 - \alpha$	$1 - \alpha < e(X)$	
Losers	82	172	5	259
Winners	4	151	82	237
All	86	323	87	496

ASSESSING UNCONFOUNDEDNESS / PLACEBO TESTS

- Main question:
  - what can we say about the plausibility of unconfoundedness?
  - Recall: unconfoundedness is not testable.
- Two Approaches: Both approaches estimate zero effects.
  - Estimate effect of treatment on pseudo outcome
  - Estimate effect of pseudo treatment on outcome

ASSESSING UNCONFOUNDEDNESS BY ESTIMATING EFFECT ON PSEUDO OUTCOME

- We consider the unconfoundedness assumption,

$$W_i \perp\!\!\!\perp \left(Y_i(\mathrm{C}), Y_i(\mathrm{T})\right) \Big| X_i. \tag{1}$$

- We partition the vector of covariates X_i into two parts, a (scalar) pseudo outcome, denoted by X_i^{p}, and the remainder, denoted by X_i^{r}, $X_i^\top = (X_i^{\mathrm{p}}, X_i^{\mathrm{r}\top})$.
- Now we assess whether the following conditional independence relation holds:

$$W_i \perp\!\!\!\perp X_i^{\mathrm{p}} \Big| X_i^{\mathrm{r}}.$$

- The two issues are
  - the interpretation of this and the relation to the unconfoundedness assumption that is of primary interest,
  - the implementation of the test.

INTERPRETATION

- The first issue concerns the link between the conditional independence relation and unconfoundedness . This link is indirect, as unconfoundedness cannot be tested directly.
- First consider an essentially stronger version of unconfoundedness:

$$W_i \perp\!\!\!\perp \left(Y_i(\mathrm{C}), Y_i(\mathrm{T})\right) \,\Big|\, X_i^{\mathrm{r}}.$$

- If this modified unconfoundedness condition were to hold, one could use the adjustment methods using only the subset of covariates X_i^{r}. In practice this is a stronger condition than the original unconfoundedness condition.
- The modified unconfoundedness condition is not testable.
- We use the pseudo outcome X_i^{p} as a proxy variable for $Y_i(\mathrm{C})$, and test

$$W_i \perp\!\!\!\perp X_i^{\mathrm{p}} \,\Big|\, X_i^{\mathrm{r}}.$$

LEADING EXAMPLE

- A leading example is that where X_i contains multiple lagged measures of the outcome.
  - In the lottery example we have observations on earnings for six, years prior to winning. Denote these lagged outcomes by $Y_{i,-1}, \ldots, Y_{i,-6}$, where $Y_{i,-1}$ is the most recent and $Y_{i,-6}$ is the most distant pre-winning earnings measure.
- One could implement the above ideas using earnings for the most recent pre-winning year $Y_{i,-1}$ as the pseudo outcome $X_i^{\mathrm{p}} = Y_{i,-1}$, so that the vector of remaining pretreatment variables $X_i^{\mathrm{r}} = (Y_{i,-2}, \ldots, Y_{i,-6})$ would still include the five prior years of pre-winning earnings $Y_{i,-2}, \ldots, Y_{i,-6}$ (ignoring additional pre-treatment variables).

IMPLEMENTATION

- Now we turn to the second issue, the implementation. One approach to testing the conditional independence assumption is to estimate the average difference in X_i^{p} by treatment status, after adjusting for differences in X_i^{r}.
- This is exactly the same problem as estimating the average effect of the treatment, using X_i^{p} as the pseudo outcome and X_i^{r} as the vector of pretreatment variables.
- We can do this using any of the methods for estimating average treatment effects.

ADDITIONAL POSSIBILITIES

- The main limitation of this approach of testing whether an adjusted average difference is equal to zero, is that it does not test all aspects of the conditional independence restriction.
- It effectively tests only whether

$$\mathbb{E}\left[\mathbb{E}\left[X_i^{\mathrm{p}}\middle| W_i = t, X_i^{\mathrm{r}}\right] - \mathbb{E}\left[X_i^{\mathrm{p}}\middle| W_i = c, X_i^{\mathrm{r}}\right]\right] = 0.$$

- The conditional independence assumption implies two additional sets of restrictions. First, of all, it implies that

$$\mathbb{E}\left[\mathbb{E}\left[g(X_i^{\mathrm{p}})\middle| W_i = t, X_i^{\mathrm{r}}\right] - \mathbb{E}\left[g(X_i^{\mathrm{p}})\middle| W_i = c, X_i^{\mathrm{r}}\right]\right] = 0,$$

  For any function $g(\cdot)$, not just the identity function.
- We can implement this by comparing average outcomes for different transformations of the pseudo outcome, and testing jointly whether any of the averages effects are zero.

ADDITIONAL POSSIBILITIES

- Second, the conditional independence restriction implies that not only on average, but in fact for all x^{r},

$$\mathbb{E}\left[X_i^{\mathrm{p}} \middle| W_i = t, X_i^{\mathrm{r}} = x^{\mathrm{r}}\right] - \mathbb{E}\left[X_i^{\mathrm{p}} \middle| W_i = c, X_i^{\mathrm{r}} = x^{\mathrm{r}}\right] = 0.$$

- One can therefore also consider tests of the form

$$\mathbb{E}\left[\mathbb{E}\left[X_i^{\mathrm{p}} \middle| W_i = t, X_i^{\mathrm{r}}\right] - \mathbb{E}\left[X_i^{\mathrm{p}} \middle| W_i = c, X_i^{\mathrm{r}}\right] \middle| X_i^{\mathrm{r}} \in \mathbb{X}_j^{\mathrm{r}}\right] = 0,$$

  for some partitioning $\{\mathbb{X}_j^{\mathrm{r}}\}_{j=1}^{J}$ of the support $\mathbb{X}^{\mathrm{r}}$ of X_i^{r}.

- That is, rather than testing whether the overall average effect of the treatment on the pseudo outcome differs from zero, one might wish to test whether the average effect in any subpopulation differs from zero.

ESTIMATING EFFECTS OF PSEUDO TREATMENTS

- Suppose we have multiple control groups.
- Let G_i be an indicator variable denoting the membership of the group.
- This indicator variable takes on three values, $G_i \in \{t, c_1, c_2\}$. For units with $G_i = c_1, c_2$, the treatment indicator W_i is equal to c:

$$W_i = \begin{cases} \mathrm{C} & \text{if } G_i = c_1, c_2, \\ \mathrm{T} & \text{if } G_i = t. \end{cases}$$

- Unconfoundedness only requires that

$$W_i \perp\!\!\!\perp \left(Y_i(\mathrm{C}), Y_i(\mathrm{T}) \right) \Big| X_i,$$

- Consider the stronger assumption

$$G_i \perp\!\!\!\perp \left(Y_i(\mathrm{C}), Y_i(\mathrm{T}) \right) \Big| X_i,$$

IMPLICATION

- This conditional independence assumption has the testable implication

$$G_i \perp\!\!\!\perp Y_i(\mathrm{C}) \;\Big|\; X_i, G_i \in \{c_1, c_2\}$$

- This is equivalent to

$$G_i \perp\!\!\!\perp Y_i^{\mathrm{obs}} \;\Big|\; X_i, G_i \in \{c_1, c_2\},$$

because $G_i \in \{c_1, c_2\}$ implies that $Y_i^{\mathrm{obs}} = Y_i(\mathrm{C})$.

IMPLEMENTATION

- Same as before. We test whether there is a difference in average values of Y_i between the two control groups, after adjusting for differences in X_i.
- That is, we effectively test whether

$$\mathbb{E}\left[\mathbb{E}\left[Y_i^{\text{obs}}\middle| G_i = c_1, X_i\right] - \mathbb{E}\left[Y_i^{\text{obs}}\middle| G_i = c_2, X_i\right]\right] = 0.$$

- We can then extend the test by simultaneously testing whether the average value of transformations of the form $g(Y_i)$ differs by group, that is, whether

$$\mathbb{E}\left[\mathbb{E}\left[g(Y_i^{\text{obs}})\middle| G_i = c_1, X_i\right] - \mathbb{E}\left[g(Y_i^{\text{obs}})\middle| G_i = c_2, X_i\right]\right] = 0.$$

- In addition we can extend the tests by testing whether, given a partition $\{\mathbb{X}_j\}_{j=1}^{J}$ of the support $\mathbb{X}$ of X_i,

$$\mathbb{E}\left[\mathbb{E}\left[Y_i^{\text{obs}}\middle| G_i = c_1, X_i\right] - \mathbb{E}\left[Y_i^{\text{obs}}\middle| G_i = c_2, X_i\right]\middle| X_i \in \mathbb{X}_j\right] = 0,$$

for all $j = 1, \ldots, J$.

ASSESSING UNCONFOUNDEDNESS USING THE LOTTERY DATA

Pseudo Outc	Covariates	Selected Cov	est	(s.e.)
Y_{-1}	$X, Y_{-6:-2}, Y_{-6:-2} > 0$	X_2, X_5, X_6, Y_{-2}	-0.53	(0.78)
$\frac{Y_{-1}+Y_{-2}}{2}$	$X, Y_{-6:-3}, Y_{-6:-3} > 0$	X_2, X_5, X_6, Y_{-3}	-1.16	(0.83)
$\frac{Y_{-1}+Y_{-2}+Y_{-3}}{3}$	$X, Y_{-6:-4}, Y_{-6:-4} > 0$	X_2, X_5, X_6, Y_{-4}	-0.39	(0.95)
$\frac{Y_{-1}+\ldots+Y_{-4}}{4}$	$X, Y_{-6:-5}, Y_{-6:-5} > 0$	X_2, X_5, X_6, Y_{-5}	-0.56	(0.97)
$\frac{Y_{-1}+\ldots+Y_{-5}}{5}$	$X, Y_{-6}, Y_{-6} > 0$	X_2, X_5, X_6, Y_{-6}	-0.41	(0.92)
$\frac{Y_{-1}+\ldots+Y_{-6}}{6}$	X	X_2, X_5, X_6	-2.56	(2.17)
Actual Outc				
Y^{obs}	$X, Y_{-1:-6}, Y_{-6:-1} > 0$	X_2, X_5, X_6, Y_{-1}	-5.74	(1.40)

ESTIMATES OF ATE ON TRANSFORMATIONS OF THE PSEUDO OUTCOME FOR SUBPOPULATIONS

Pseudo Outcome	Subpopulation	est	(s.e.)
$1\{Y_{-1} = 0\}$	$Y_{-2} = 0$	-0.07	(0.78)
$1\{Y_{-1} = 0\}$	$Y_{-2} > 0$	0.02	(0.02)
Y_{-1}	$Y_{-2} = 0$	-0.31	(0.30)
Y_{-1}	$Y_{-2} > 0$	0.05	(0.06)
		statistic	p-value
Combined Statistic (chi-squared, dof 4)		2.20	0.135

Lalonde Data: Normalized Diffs in Covariates after Subclassification

	Full	Matched	2 Subclasses	5 Subclasses
Black	2.43	0.00	0.00	0.00
Hispanic	-0.05	0.00	0.00	0.00
Age	-0.80	-0.15	0.13	0.09
Married	-1.23	-0.28	0.18	0.02
Nodegree	0.90	0.25	-0.21	-0.06
Education	-0.68	-0.18	0.16	0.05
E'74	-1.57	-0.03	-0.04	-0.05
U'74	1.49	0.02	0.09	0.07
E'75	-1.75	-0.07	-0.03	-0.02
U'75	1.19	0.02	0.13	0.07

THE LALONDE DATA

- Next let us look at the estimates for effect on 1978 earnings.
- We use the matched sample and use subclassification to remove additional bias.

	Full Sample		Matched		Matched		Matched	
Cov	1 Block		1 Block		2 Blocks		4 Blocks	
No	-8.50	0.58	1.72	0.74	1.81	0.75	1.86	0.76
Few	0.69	0.59	1.81	0.73	1.80	0.73	1.99	0.75
All	1.07	0.55	1.97	0.66	1.90	0.67	2.06	0.66

THE LALONDE DATA

- Covariate still make some difference, but reasonably robust.
- With all covariates included the estimate is close to experimental one, but would we have known that?
- With only the trainee and CPS comparison data would we have concluded that the evaluation was succesful?
- Analyse trainee/CPS-control data with earnings '75 as outcome, and remaining eight covariates as pre-treatment variables.
- We take the original full sample, create a matched sample basedon the eight remaining covariates, and estimate the "effect" of the training on 1975 earnings.
- Also: test effect on earnings '75, earnings '75>0, separately for those with positive and zero earnings in 1974. Leads to 4 estimated effects, and we test that all four are jointly zero.

THE LALONDE DATA: ESTIMATES FOR PSEUDO OUTCOME E'75

			p-value
earnings 1975:	-0.90	(0.33)	0.006
chi-squared test	53.8	(dof=4)	< 0.001

- 0.95 quantile of a chi-squared distribution with degrees of freedom equal to four is 9.49, much smaller than 53.8.
- These estimates differ from zero, so estimates for 1978 are not necessarily credible.
- In the Lalonde case, had we done this analysis first, we would not necessarily have even estimated the effect for 1978 earnings: the initial analysis suggests that the estimates would not necessarily be credible.

ASSESSING UNCONFOUNDEDNESS: SUMMARY

- Whenever possible, assess the plausibility of unconfoundedness.
- Having good covariates available is important here.
- Analysis may suggest that results are not credible. They dont point in direction of alternative analysis, only suggest that proceeding under the assumption of unconfoundedness is not reasonable.
- for the lottery data we obtain credible and precise estimates, robust to small changes in the specification and with unconfoundedness plausible.
- for the lalonde data the results are more mixed. the estimates are robust to changes in the specification, but the assessments of unconfoundedness raises doubts.

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